AP CALCULUS AB

Unit 1 • Self Check 1.2 • Lessons 1-3 and 1-4 • Answer key

Self Check 1.2 • Lessons 1-3 and 1-4

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$$\lim_{x\to\infty} \frac{3x^2+1}{x^2-5} =$$

(A) 3

(B) $-\frac{1}{5}$

(C) 0

(D) ∞

Solution

Answer: (A).

Question 2

NO CALCULATOR

$$\lim_{x\to 2} \frac{3}{(x-2)^2} =$$

(A) ∞

(B) $-\infty$

(C) 3

(D) It does not exist and is not infinite.

Solution

Answer: (A).

Question 3

NO CALCULATOR

For what a is $f(x) = \begin{cases} ax+2, & x<3 \\ x^2-a, & x\geq 3 \end{cases}$ continuous?

(A) $\frac{11}{4}$

(B) $\frac{7}{4}$

(C) $\frac{7}{2}$

(D) 7

Solution

Answer: (B). $3a+2=9-a$.

Question 4

NO CALCULATOR

Which condition for continuity fails for $f(x) = \frac{x^2-9}{x-3}$ at $x = 3$?

(A) $f(3)$ is defined
(B) the limit exists
(C) the limit equals $f(3)$
(D) none

Solution
Answer: (A). $f(3)$ is undefined.

Question 5 (Free response)

NO CALCULATOR

Let $f(x) = \dfrac{2x^2 - 8}{x^2 - x - 2}$. Find the vertical asymptote, the hole and the horizontal asymptote. Then classify the discontinuities of $g(x) = \frac{x+2}{x^2-4}$.

Solution
$f = \frac{2(x-2)(x+2)}{(x-2)(x+1)}$: VA $x = -1$; hole at $x = 2$, height $\frac{2(4)}{3} = \frac{8}{3}$; HA $y = 2$. g: removable at $x = -2$ (limit $-\frac{1}{4}$), infinite at $x = 2$.